



Enantioselective copper(II)-catalyzed conjugate addition of diethylzinc to β -substituted enones utilizing BINOL-based phosphoramidite ligands

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ABSTRACT

A family of new chiral phosphoramidite ligands based on 3,3'-disubstituted (S)-BINOL have been successfully synthesized. The chiral phosphoramidite ligands were subsequently applied to the copper(II)-catalyzed enantioselective addition of diethylzinc to β -substituted enones, giving the desired chiral adducts with enantioselectivities of up to 93% ee and high yields (up to 88%).

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1. Introduction

The synthesis of enantiomerically pure compounds has been well studied by chemists over the past several decades and such compounds have been widely utilized in organic synthesis, pharmaceutical, food and the related areas.¹ As a result, the development of known or new chiral catalysts with excellent catalytic activities and stereoselectivities still constitutes one of the most challenging endeavors in the field of asymmetric synthesis. The enantioselective addition of diethylzinc to carbonyl compounds (a 1,2-addition) and to β -substituted enones (a 1,4-addition) is an attractive method for creating a stereogenic center in an organic molecule.² In reported 1,4-addition reaction of β -substituted enones using dialkylzinc reagents, chiral copper and nickel complexes are the most widely developed catalytic systems for inducing high stereoselectivities.³ In the case of copper, a variety of diphosphine,⁴ diphosphite,⁵ spiro-phosphoramidites,⁶ aminophosphine⁷ and other ligands⁸ have been developed for this transformation. In particular, BINOL-derived phosphoramidites have been shown to have significantly high stereocontrol in the ZnEt_2 addition to β -substituted enones.⁹ Consequently, phosphine-containing ligands have recently received much attention in this area. Very recently, BINOL or TADDOL-derived phosphoramidites have proven to be excellent chiral ligands for the palladium-catalyzed intramolecular C–H arylation reaction of *N*-(2-bromoaryl)ferrocenecarboxamides^{10a} and *N*-(2-haloaryl)-*P,P*-diphenylphosphinic amides^{10b} by us, giving planar chiral ferrocene derivatives and *P*-stereogenic phosphine oxide derivatives in good yields and with high stereoselectivities. In

addition, we have also reported on a highly enantioselective asymmetric addition of diethyl zinc to aldehydes catalyzed by *N*-ferrocenylmethyl aziridino alcohol ligands bearing Fréchet-type dendrimers.¹¹ The ligands could be recycled and reused eight times for such a reaction with almost the original high reactivity and selectivity. On the other hand, Feringa et al.¹² and Buono et al.¹³ have demonstrated that the introduction of substituents at the 3- and 3'-positions of the binaphthol part of the phosphoramidite ligands has a positive effect on the enantioselectivity. For example, the applications of chiral phosphoramidites based on 3,3'-disubstituted (R)-BINOL in cobalt-catalyzed [6+2] cycloadditions of cycloheptatriene with alkynes, the ees were very sensitive to substituents at the 3- and 3'-positions of the BINOL moiety.¹³ Encouraged by the above results and also in continuation of our efforts to develop chiral ligands for asymmetric synthesis, we have decided to design a series of novel chiral phosphoramidite ligands based on 3,3'-disubstituted (S)-BINOL and their application in copper(II)-catalyzed enantioselective addition of diethylzinc to β -substituted enones. The addition reactions afforded the desired chiral adducts with good to excellent enantioselectivities (up to 93% ee).

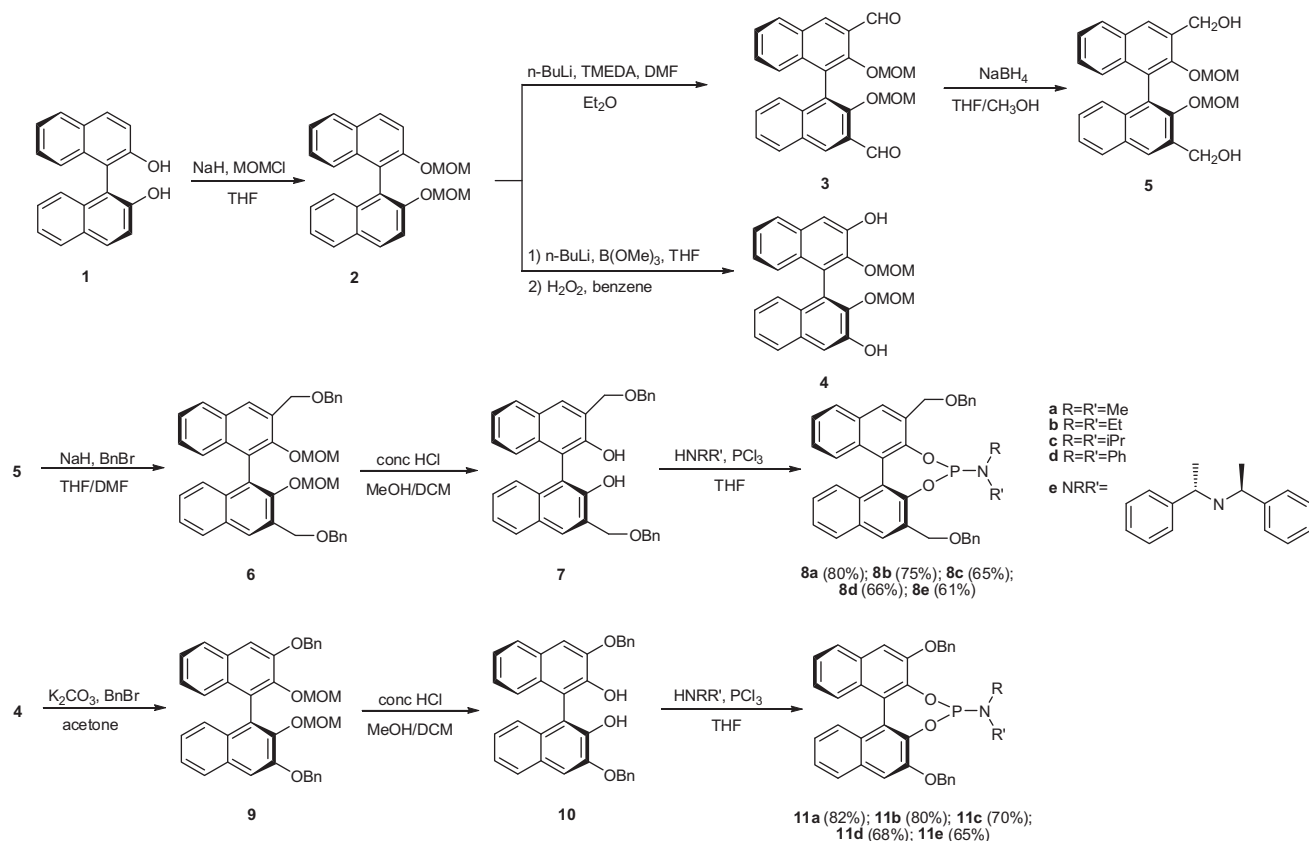
2. Results and discussion

2.1. Synthesis of phosphoramidite ligands

The synthesis of the required chiral ligands **8a–e** was easily carried out in a six step sequence from commercially available (S)-BINOL as shown in Scheme 1. The key BINOL derivative 3,3'-bis((benzyloxy)methyl)-[1,1'-binaphthalene]-2,2'-diol **7** was synthesized using the procedure established by Fan et al.¹⁴ Compound **7** was readily converted to the corresponding phosphoramidite

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Scheme 1. Synthesis of BINOL-based phosphoramidite ligands.

ligands **8a–e** by reacting with an excess of amine and phosphorus trichloride in THF. The expected phosphoramidite ligands **8a–e** were isolated in good yields (61–80%) after purification. For the sake of comparison, compounds **11a–e** were also synthesized using the same method (as shown in Scheme 1). All of the new compounds were fully characterized by ^1H NMR, ^{13}C NMR, ^{31}P NMR, HRMS, and IR spectra.

2.2. Catalytic study

With the BINOL-based phosphoramidite ligands **8a–e** and **11a–e** in hand, their potential in the copper(II)-catalyzed enantioselective addition of diethylzinc to β -substituted enones was then evaluated. Our study began with the asymmetric addition of diethylzinc to chalcone **12a** as the model reaction, which was carried out in the presence of chiral phosphoramidite (5.0 mol %) and $\text{Cu}(\text{OTf})_2$ (2.5 mol %) in toluene at -40°C for 48 h. Under the stated conditions, all of the chiral phosphoramidites **8a–e** and **11a–e** afforded the expected 1,3-diphenylpentan-1-one **13a** in good yields (65–85%, Table 1, entries 1–10). Chiral phosphoramidite **8c** gave relatively higher enantioselectivity, which proved that bulkier *N*-substituents led to better enantioselectivities (entry 3 vs entries 1–2 and 4). Unfortunately, the introduction of the substituents at 3- and 3'-positions had almost no significant effect (Table 1, entry 3 vs Table 3, entry 1 in Ref. 12). Efforts to increase the enantioselectivity by the introduction of a chiral amine moiety to the ligand were unsuccessful (entry 5). Compounds **11a–e** gave inferior results than their counterparts **8a–e**, which indicated that both the electronic and steric effects of the BINOL-moiety were crucial for the ligands. The reduction of the catalyst loading from 5.0 mol % to 2.5 mol % led to a decreased enantioselectivity (79% ee, entry 11). When the reaction was conducted with 7.5 mol % of **8c**, the enantioselectivity

Table 1
Reaction condition optimization^a

Entry	L*	Solvent	Temp ($^\circ\text{C}$)	Yield ^b (%)	ee ^{c,d} (%)
1	8a (5 mol %)	Toluene	-40	74	57
2	8b (5 mol %)	Toluene	-40	76	70
3	8c (5 mol %)	Toluene	-40	85	87
4	8d (5 mol %)	Toluene	-40	80	49
5	8e (5 mol %)	Toluene	-40	65	58
6	11a (5 mol %)	Toluene	-40	84	0
7	11b (5 mol %)	Toluene	-40	76	5
8	11c (5 mol %)	Toluene	-40	72	53
9	11d (5 mol %)	Toluene	-40	80	30
10	11e (5 mol %)	Toluene	-40	70	68
11	8c (2.5 mol %)	Toluene	-40	87	79
12	8c (7.5 mol %)	Toluene	-40	88	82
13	8c (5 mol %)	Toluene	-60	86	84
14	8c (5 mol %)	Toluene	-78	29	32
15	8c (5 mol %)	THF	-40	35	47
16	8c (5 mol %)	CH_2Cl_2	-40	82	61
17	8c (5 mol %)	Et_2O	-40	63	23

^a The reaction was run with chalcone (0.20 mmol), Et_2Zn (0.40 mmol, 1 M in hexane), $\text{Cu}(\text{OTf})_2$ (2.5 mol %), solvent (2 mL) for 48 h.

^b Isolated yields.

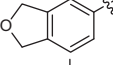
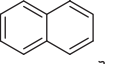
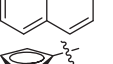
^c Enantiomeric excess was determined by chiral HPLC analysis.

^d The absolute configurations of the products were assigned to be (*R*) by comparison of specific rotations with those in the literature.^{8c}

was not enhanced further (entry 12). When the temperature was lowered to -60°C or -78°C , the ee value did not improve (entries 13 and 14). Several other solvents including THF, CH_2Cl_2 , and Et_2O

Table 2

Asymmetric addition of diethylzinc to β -substituted enones catalyzed by the copper complex of phosphoramidite **8c**^a

$\text{R}^1\text{-CH=CH-C(=O)R}^2 + \text{Et}_2\text{Zn} \xrightarrow[\text{toluene, -40}^\circ\text{C}]{\text{ligand } \mathbf{8c}, \text{Cu(OTf)}_2} \text{R}^1\text{-CH}_2\text{-CH}_2\text{-C(=O)R}^2$					
Entry	R ¹	R ²	Product	Yield ^b (%)	ee ^{c,d} (%)
1	<i>p</i> -MeOC ₆ H ₄	Ph	13b	82	67
2	<i>m</i> -MeOC ₆ H ₄	Ph	13c	68	79
3	<i>o</i> -MeOC ₆ H ₄	Ph	13d	78	72
4	<i>p</i> -MeC ₆ H ₄	Ph	13e	81	80
5	<i>p</i> -FC ₆ H ₄	Ph	13f	83	87
6	<i>p</i> -ClC ₆ H ₄	Ph	13g	76	88
7	<i>o</i> -ClC ₆ H ₄	Ph	13h	78	58
8	<i>p</i> -CF ₃ C ₆ H ₄	Ph	13i	77	83
9		Ph	13j	70	62
10		Ph	13k	75	93
11		Ph	13l	73	73
12		Ph	13m	30	71 ^e
13	Ph	<i>p</i> -ClC ₆ H ₄	13n	61	82
14	Ph	<i>p</i> -MeOC ₆ H ₄	13o	81	65
15	Ph		13p	20	53
16	<i>p</i> -ClC ₆ H ₄	<i>p</i> -ClC ₆ H ₄	13q	72	91
17	<i>p</i> -MeOC ₆ H ₄	<i>p</i> -ClC ₆ H ₄	13r	82	71
18		<i>p</i> -ClC ₆ H ₄	13s	42	87 ^e
19	<i>p</i> -ClC ₆ H ₄	<i>p</i> -MeOC ₆ H ₄	13t	64	84
20	<i>p</i> -MeOC ₆ H ₄	<i>p</i> -MeOC ₆ H ₄	13u	20	39
21		<i>p</i> -MeOC ₆ H ₄	13v	20	39 ^e

^a The reaction was run with β -substituted enones (0.20 mmol) and Et₂Zn (0.40 mmol, 1 M in hexane) in the presence of ligand **8c** (5.0 mol %) and Cu(OTf)₂ (2.5 mol %) in 2 mL of toluene at -40°C for 48 h.

^b Isolated yields.

^c Enantiomeric excess was determined by chiral HPLC analysis.

^d The absolute configurations of the products were assigned to be (R) by comparison of the specific rotations with those in the literature.^{8c}

^e The absolute configuration of the products were assigned to be S by comparison of optical rotation with that in Ref. **8c**.

were also tested at -40°C and none of them gave better enantioselectivity than toluene (entry 3 vs entries 15–17). Overall, the optimized reaction conditions were established as the following: ligand **8c** (5.0 mol %), Cu(OTf)₂ (2.5 mol %), toluene as the solvent, at -40°C for 48 h. In addition, BINOL-based phosphoramidite possessing 3-phenylpropyl at 3- and 3'-positions has also been prepared. The chiral phosphoramidite was applied to the copper (II)-catalyzed enantioselective addition of diethylzinc to chalcone, and the expected adducts were obtained with relatively low ee (54% ee) (see in Supporting information). The results mean that the introduction of an oxygen atom at 3- and 3'-positions has a positive effect on the conjugate addition reaction.

Under the above optimized reaction conditions, various β -substituted enones reacted smoothly with diethylzinc to give the desired product in good chemical yields and enantioselectivities (Table 2). In

comparison to chalcone with a β -phenyl group, the enantioselectivities decreased slightly when the β -aryl substituents bare electron-donating groups such as *p*-MeO, *m*-MeO, and *o*-MeO (Table 2, entries 1–3) or an electron neutral group such as *p*-Me (entry 4). For the substrates with electron-withdrawing groups such as *p*-F, *p*-Cl, or *p*-CF₃ on the β -aryl substituents, comparable enantioselectivities were achieved (entries 5, 6 and 8). The position of the group on the β -aryl ring also had some influence on the enantioselectivity. For example, the reactions of (*E*)-3-(2-chlorophenyl)-1-phenylprop-2-en-1-one with diethylzinc afforded the expected adducts with relatively low ee value (58% ee, entry 7). When the β -aryl substituent was a 1-naphthyl group, excellent stereocontrol of the corresponding product was observed (93% ee, entry 10). The ee values were slightly lower when the β -aryl substituent was a 2-naphthyl group (73% ee, entries 11). Similarly, (*E*)-3-(1,3-dihydroisobenzofuran-5-yl)-1-phenylprop-2-en-1-one and (*E*)-3-(ferrocenyl)-1-phenylprop-2-en-1-one which are more electron-rich than (*E*)-chalcone, also afforded lower stereoselectivities (62% and 71% ee, entries 9 and 12). In addition, the effect of the electron-withdrawing group in enhancing the enantioselectivities was also found in the case of enone substrates bearing a *p*-ClC₆H₄ moiety attached to the carbonyl group (entry 13 vs entries 14 and 15). Finally, the asymmetric addition of diethylzinc to other β -substituted enones was investigated with **8c** as the ligand (entries 16–21). Overall, good to excellent enantioselectivities were achieved (entries 16–19). When both of the aryl rings were substituted with electron-rich groups, the yield and enantioselectivity decreased dramatically (entries 20 and 21). In addition, the asymmetric conjugate addition of diethylzinc to cyclohex-2-enone was evaluated. When using the ligand **8c** as the catalyst, only trace amounts of the product were obtained (data not shown in Table 2).

3. Conclusion

In conclusion, a novel class of phosphoramidite ligands based on 3,3'-disubstituted (S)-BINOL as chiral catalysts has been developed. These compounds are easy to synthesize and their structures can be readily modified. Ligand **8c** was coordinated to Cu(II) and then utilized as a catalyst for the enantioselective addition of diethylzinc to various β -substituted enones, giving the desired chiral adducts in high yields with enantioselectivities of up to 93% ee. Further modification of the phosphoramidite ligands and their applications in asymmetric catalysis are currently in progress.

4. Experimental

4.1. General

Solvents were dried with standard methods and freshly distilled prior to use if needed. All other chemicals were used as purchased. Melting points were determined using YRT-3 melting point apparatus and are uncorrected. Infrared spectra were obtained with a Bruker VECTOR 22 spectrophotometer in KBr pellets. ¹H, ¹³C and ³¹P{¹H} NMR spectra were recorded on a Bruker DPX-400 spectrometer in CDCl₃ with TMS as an internal standard for ¹H, ¹³C NMR and 85% H₃PO₄ as the external standard for ³¹P{¹H} NMR. HRMS were determined on a Waters Q-ToF Micro MS/MS System ESI spectrometer. Optical rotations were recorded on a Perkin Elmer 341 polarimeter. Compounds **7** and **10** were synthesized according to similar literature methods.¹⁵

4.2. General procedure for the synthesis of compounds **8a–b** and **11a–b**

Ligand **7** (263.3 mg, 0.50 mmol) [or **10** (249.3 mg, 0.50 mmol)] was dissolved in 10 mL of anhydrous toluene and cooled to 0°C .

After the addition of $P(NMe_2)_3$ or $P(NEt_2)_3$ (0.65 mmol), the resulting solution mixture was stirred at 0 °C for 10 min. Next, the reaction mixture was gently warmed to 110 °C for 6 h, and then filtered, concentrated, and purified by preparative TLC on silica gel plates to give the pure product **8a–b** [or **11a–b**].

Characterization data of new ligand 8a: Yield: 80%; mp: 46–48 °C. $[\alpha]_D^{20} = +352$ (c 0.2, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 8.12 (s, 2H), 7.92 (dd, $J = 18.9$, 13.6 Hz, 2H), 7.50 (t, $J = 8.5$ Hz, 4H), 7.44–7.21 (m, 12H), 4.90 (dd, $J = 27.3$, 13.3 Hz, 4H), 4.80–4.68 (m, 4H), 2.43–2.28 (m, 6H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 147.6, 147.5, 138.4, 138.3, 132.4, 132.0, 131.0, 130.6, 130.5, 130.3, 128.6, 128.5, 128.4, 128.3, 127.8, 127.7, 127.6, 127.5, 126.9, 126.8, 125.8, 125.7, 124.9, 124.7, 124.0, 123.9, 122.7, 73.1, 72.8, 68.12, 68.10, 67.7, 35.6, 35.4. ^{31}P NMR (162 MHz, $CDCl_3$) δ 148.62. IR(KBr) ν 3056, 2968, 2963, 2857, 1450, 1350, 1108, 1018, 900, 742, 965 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{38}H_{34}NO_4P$ $[M+H]^+$: 600.2304 found: 600.2306.

Ligand 8b: Yield: 75%; mp: 44–46 °C. $[\alpha]_D^{20} = +363$ (c 0.2, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 8.09 (d, $J = 5.6$ Hz, 2H), 7.93–7.90 (m, 2H), 7.47 (t, $J = 6.5$ Hz, 4H), 7.43–7.35 (m, 6H), 7.33–7.18 (m, 6H), 4.94 (d, $J = 13.5$ Hz, 3H), 4.82–4.72 (m, 5H), 2.97–2.76 (m, 4H), 0.93 (t, $J = 7.0$ Hz, 6H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 148.4, 148.3, 148.1, 138.8, 138.7, 132.8, 132.4, 131.4, 131.1, 130.9, 129.1, 128.9, 128.8, 128.7, 128.24, 128.20, 128.1, 127.7, 127.4, 126.2, 126.0, 125.3, 125.0, 124.5, 124.4, 122.6, 73.5, 73.3, 68.6, 68.3, 39.4, 39.1, 15.3. ^{31}P NMR (162 MHz, $CDCl_3$) δ 158.78. IR(KBr) ν 3030, 2969, 2922, 2857, 1450, 1349, 1108, 900, 742 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{40}H_{38}NO_4P$ $[M+H]^+$: 628.2617 found: 628.2619.

Ligand 11a: Yield: 82%; mp: 91–92 °C. $[\alpha]_D^{20} = +415$ (c 0.1, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 7.79 (dd, $J = 18.7$, 8.1 Hz, 2H), 7.60 (dd, $J = 15.2$, 7.6 Hz, 4H), 7.48–7.27 (m, 12H), 7.19–7.10 (m, 2H), 5.42 (s, 2H), 5.34 (dd, $J = 31.4$, 11.8 Hz, 2H), 2.53 (d, $J = 9.3$ Hz, 6H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 160.2, 160.1, 148.0, 147.9, 147.7, 140.93, 140.92, 136.94, 136.90, 132.4, 132.1, 131.0, 130.6, 130.44, 130.42, 128.7, 128.6, 128.3, 128.2, 128.0, 127.9, 127.6, 127.5, 127.4, 127.0, 126.9, 125.8, 125.7, 124.9, 124.6, 124.0, 123.9, 122.2, 122.1, 106.7, 106.6, 101.5, 101.4, 72.9, 72.8, 70.1, 68.2, 68.1, 67.9, 39.0, 38.8, 15.0, 14.9. ^{31}P NMR (162 MHz, $CDCl_3$) δ 149.30. IR(KBr) ν 3030, 2883, 1592, 1442, 1245, 973, 911, 742, 690 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{36}H_{30}NO_4P$ $[M+H]^+$: 572.1985 found: 572.1987.

Ligand 11b: Yield: 80%; mp: 78–80 °C. $[\alpha]_D^{20} = +366$ (c 0.1, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 7.75 (t, $J = 8.5$ Hz, 2H), 7.55–7.53 (m, 4H), 7.41–7.30 (m, 11H), 7.25–7.22 (m, 1H), 7.14–7.06 (m, 2H), 5.39–5.22 (m, 4H), 3.12–3.01 (m, 2H), 2.96–2.84 (m, 2H), 0.94 (t, $J = 7.1$ Hz, 6H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 150.9, 150.8, 143.11, 143.06, 142.1, 137.5, 137.3, 132.0, 131.2, 128.9, 128.4, 128.3, 128.2, 128.1, 127.9, 127.5, 127.4, 127.3, 126.3, 126.2, 125.6, 125.4, 124.4, 124.2, 110.4, 109.6, 71.1, 70.9, 39.0, 38.7, 15.0, 14.9. ^{31}P NMR (162 MHz, $CDCl_3$) δ 149.45. IR(KBr) ν 3058, 2971, 2926, 2870, 1594, 1442, 1246, 1109, 911, 743 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{38}H_{34}NO_4P$ $[M+H]^+$: 622.2123 found: 622.2124.

4.3. General procedure for the synthesis of compounds 8c–e and 11c–e

Amine (0.80 mmol) was dissolved in 4 mL of anhydrous THF and cooled to –78 °C. After the addition of *n*-BuLi (0.80 mmol, 0.32 mL, 2.5 M in hexane), the resulting solution mixture was stirred for 30 min, after which PCl_3 (0.8 mmol, 70 μ L) was added to the reaction mixture. The reaction mixture was then gently warmed to 0 °C for 1 h, after which Et_3N (0.45 mL, 3.2 mmol) and ligand **7** (263.3 mg, 0.50 mmol) [or **10** (249.3 mg, 0.50 mmol)] were added to the reaction mixture and stirred at room temperature for 12 h. After cooling, filtration, and evaporation, the residue was purified by preparative TLC on silica gel plates to give the pure product **8c–e** or **11c–e**.

Ligand 8c: Yield: 65%; mp: 45–46 °C. $[\alpha]_D^{20} = +356$ (c 0.1, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 8.10 (d, $J = 9.1$ Hz, 2H), 7.91 (t, $J = 8.4$ Hz, 2H), 7.46 (d, $J = 7.4$ Hz, 4H), 7.42–7.29 (m, 9H), 7.24–7.16 (m, 3H), 4.97–4.89 (m, 4H), 4.83–4.70 (m, 4H), 3.32–3.23 (m, 2H), 1.27–0.82 (m, 12H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 149.5, 149.4, 149.3, 139.7, 139.6, 133.7, 133.5, 132.3, 132.0, 131.7, 131.6, 129.9, 129.8, 129.7, 129.6, 129.5, 129.24, 129.21, 129.1, 129.0, 128.8, 128.4, 128.2, 127.1, 126.8, 126.1, 125.8, 125.2, 125.1, 123.2, 123.1, 74.5, 74.4, 69.5, 69.4, 69.3, 46.5, 26.2. ^{31}P NMR (162 MHz, $CDCl_3$) δ 152.58. IR(KBr) ν 3033, 2965, 2857, 1452, 1354, 1110, 1071, 902, 743 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{42}H_{42}NO_4P$ $[M+Na]^+$: 678.2749 found: 678.2748.

Ligand 8d: Yield: 66%; mp: 44–45 °C. $[\alpha]_D^{20} = +107$ (c 0.1, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 8.09 (s, 1H), 7.90 (d, $J = 8.2$ Hz, 1H), 7.78 (d, $J = 5.7$ Hz, 2H), 7.42–7.25 (m, 12H), 7.19–7.16 (m, 4H), 6.97–6.86 (m, 10H), 4.96–4.87 (m, 3H), 4.72–4.52 (m, 5H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 145.8, 143.4, 143.3, 137.2, 137.1, 131.4, 130.9, 130.1, 129.6, 129.1, 128.0, 127.6, 127.4, 127.3, 127.2, 126.86, 126.82, 126.7, 126.6, 125.9, 125.0, 124.8, 124.7, 124.6, 124.1, 123.6, 123.3, 72.0, 71.8, 67.1, 66.7. ^{31}P NMR (162 MHz, $CDCl_3$) δ 140.34. IR(KBr) ν 3028, 2850, 1489, 1348, 1241, 957, 900, 743, 689, 519 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{48}H_{38}NO_4P$ $[M+H]^+$: 724.2617 found: 724.2615.

Ligand 8e: Yield: 61%; mp: 53–54 °C. $[\alpha]_D^{20} = +27.1$ (c 0.1, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 8.10 (s, 2H), 7.89 (t, $J = 8.3$ Hz, 2H), 7.50 (d, $J = 7.5$ Hz, 2H), 7.41–7.28 (m, 11H), 7.20–7.12 (m, 9H), 6.96–6.94 (m, 4H), 5.04–4.97 (m, 2H), 4.79 (s, 2H), 4.54–4.47 (m, 2H), 4.36–4.20 (m, 3H), 4.00 (d, $J = 14.4$ Hz, 1H), 1.60 (d, $J = 6.6$ Hz, 6H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 148.8, 148.7, 139.7, 133.6, 133.5, 132.5, 131.8, 131.7, 131.6, 129.9, 129.8, 129.6, 129.5, 129.3, 129.2, 129.1, 129.0, 128.9, 128.3, 128.2, 128.1, 127.1, 126.2, 126.0, 125.2, 125.1, 123.3, 74.6, 74.3, 69.5, 69.3, 69.2, 53.4, 53.3. ^{31}P NMR (162 MHz, $CDCl_3$) δ 145.46. IR(KBr) ν 3028, 2922, 2855, 1447, 1353, 1108, 1073, 901, 714, 695 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{52}H_{46}NO_4P$ $[M+H]^+$: 780.3246 found: 780.3243.

Ligand 11c: Yield: 70%; mp: 71–72 °C. $[\alpha]_D^{20} = +248$ (c 0.1, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 7.82–7.79 (m, 2H), 7.59 (t, $J = 7.2$ Hz, 4H), 7.44–7.25 (m, 10H), 7.24–7.13 (m, 3H), 7.10 (t, $J = 7.6$ Hz, 1H), 5.34 (s, 2H), 5.26 (dd, $J = 30.2$, 11.6 Hz, 2H), 3.49–3.34 (m, 2H), 1.14 (dd, $J = 18.1$, 6.7 Hz, 12H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 150.6, 150.5, 146.4, 143.8, 142.3, 137.2, 136.7, 136.0, 131.6, 130.6, 129.0, 128.8, 128.6, 128.58, 128.47, 128.40, 128.35, 127.9, 127.8, 127.7, 127.6, 127.2, 127.0, 126.9, 126.8, 125.1, 124.8, 124.7, 124.1, 123.7, 123.6, 109.7, 109.0, 107.6, 71.1, 70.4, 70.3, 44.9, 44.8, 24.8, 24.7, 23.8, 23.7. ^{31}P NMR (162 MHz, $CDCl_3$) δ 152.11. IR(KBr) ν 3058, 2963, 2886, 1593, 1444, 1248, 1163, 1112, 913, 742 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{40}H_{38}NO_4P$ $[M+H]^+$: 650.2436 found: 650.2438.

Ligand 11d: Yield: 68%; mp: 74–76 °C. $[\alpha]_D^{20} = +189$ (c 0.1, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 7.75 (d, $J = 8.1$ Hz, 1H), 7.62 (d, $J = 8.2$ Hz, 1H), 7.54–7.51 (m, 4H), 7.43–7.31 (m, 10H), 7.25–7.18 (m, 2H), 7.10–7.08 (m, 6H), 6.95–6.90 (m, 6H), 5.41–5.34 (m, 2H), 5.09 (d, $J = 11.3$ Hz, 1H), 4.93 (d, $J = 11.4$ Hz, 1H). ^{13}C NMR (100 MHz, $CDCl_3$) δ 151.42, 151.40, 151.2, 146.04, 145.94, 143.4, 143.3, 142.0, 138.2, 137.9, 133.0, 132.4, 129.89, 129.87, 129.6, 129.3, 129.2, 129.1, 128.43, 128.37, 128.29, 128.2, 128.1, 128.0, 126.7, 126.2, 125.7, 125.2, 124.8, 111.3, 109.5, 72.0, 71.2. ^{31}P NMR (162 MHz, $CDCl_3$) δ 140.44. IR(KBr) ν 3058, 2923, 1591, 1443, 1246, 1159, 1109, 914, 745, 695 cm^{-1} . HRMS (ESI) m/z : Calcd for $C_{46}H_{34}NO_4P$ $[M+Na]^+$: 718.2123 found: 718.2125.

Ligand 11e: Yield: 65%; mp: 81–83 °C. $[\alpha]_D^{20} = +78$ (c 0.1, $CHCl_3$). 1H NMR (400 MHz, $CDCl_3$) δ 7.78 (d, $J = 8.1$ Hz, 1H), 7.70 (d, $J = 8.3$ Hz, 1H), 7.61 (d, $J = 7.3$ Hz, 2H), 7.47 (d, $J = 7.2$ Hz, 2H), 7.42–7.27 (m, 10H), 7.23 (d, $J = 9.9$ Hz, 4H), 7.15–7.00 (m, 6H), 6.92 (t, $J = 7.6$ Hz, 4H), 5.34 (s, 2H), 5.20 (d, $J = 12.1$ Hz, 1H), 5.07

(d, $J = 12.1$ Hz, 1H), 4.40–4.28 (m, 2H), 1.57 (d, $J = 7.0$ Hz, 6H). ^{13}C NMR (100 MHz, CDCl_3) δ 149.2, 142.2, 140.8, 135.83, 135.78, 130.6, 129.8, 127.5, 127.4, 126.8, 126.7, 126.6, 126.5, 126.4, 126.2, 126.1, 126.0, 125.8, 125.3, 124.2, 123.8, 122.7, 122.6, 108.1, 69.4, 69.1, 54.3, 54.1, 22.3, 22.2. ^{31}P NMR (162 MHz, CDCl_3) δ 143.00. IR (KBr) ν 3030, 2922, 1442, 1245, 1111, 912, 744, 695 cm^{-1} . HRMS (ESI) m/z : Calcd for $\text{C}_{50}\text{H}_{42}\text{NO}_4\text{P}$ $[\text{M}+\text{Na}]^+$: 774.2749 found: 774.2747.

4.4. General procedure for the catalytic asymmetric addition of diethylzinc to β -substituted enones

A solution of $\text{Cu}(\text{OTf})_2$ (1.8 mg, 0.005 mmol) and ligand **8** or **11** (0.01 mmol) in 1.5 mL toluene was stirred for 2.0 h. The solution was then cooled to -40°C and β -substituted enones (0.2 mmol) and a 1.0 M hexane solution of diethylzinc (0.4 mL, 0.5 mmol) were added. After stirring for 48 h at -40°C , the solution was mixed with saturated ammonium chloride solution, and the product was extracted with ethyl acetate. The residue was purified by preparative TLC on silica gel plates to afford the desired products. The enantiomeric excess was determined using HPLC with a Chiralpak AD-H chiral column (250 mm $\times$ 4.6 mm). In all cases, the chromatograms of the products were compared against a known racemic mixture.

(*R*)-1,3-Diphenylpentan-1-one **13a**:^{16,18} The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 11.3 min, 13.9 min (major), 87% ee.

(*R*)-3-(4-Methoxyphenyl)-1-phenylpentan-1-one **13b**:^{16–18} The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 14.8 min, 21.7 min (major), 67% ee.

(*R*)-3-(3-Methoxyphenyl)-1-phenylpentan-1-one **13c**:¹⁷ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 13.3 min, 15.7 min (major), 79% ee.

(*R*)-3-(2-Methoxyphenyl)-1-phenylpentan-1-one **13d**:¹⁷ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 12.2 min, 14.9 min (major), 72% ee.

(*R*)-1-Phenyl-3-(*p*-tolyl)pentan-1-one **13e**:^{17,19} The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 10.5 min, 14.3 min (major), 80% ee.

(*R*)-3-(4-Fluorophenyl)-1-phenylpentan-1-one **13f**:^{18,19} The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 11.6 min, 14.9 min (major), 87% ee.

(*R*)-3-(4-Chlorophenyl)-1-phenylpentan-1-one **13g**:¹⁷ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 12.0 min, 16.6 min (major), 88% ee.

(*R*)-3-(2-Chlorophenyl)-1-phenylpentan-1-one **13h**:¹⁷ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 27.8 min, 33.6 min (major), 58% ee.

(*R*)-1-Phenyl-3-(4-(trifluoromethyl)phenyl)pentan-1-one **13i**:^{17,19} The enantiomeric excess was determined on a Daicel Chiralpak

AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 9.8 min, 12.7 min (major), 83% ee.

(*R*)-3-(1,3-Dihydroisobenzofuran-5-yl)-1-phenylpentan-1-one **13j**:¹⁶ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 27.6 min, 37.4 min (major), 62% ee.

(*R*)-3-(Naphthalen-1-yl)-1-phenylpentan-1-one **13k**:¹⁹ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 23.9 min, 29.8 min (major), 93% ee.

(*R*)-3-(Naphthalen-2-yl)-1-phenylpentan-1-one **13l**:^{17,19} The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 21.1 min, 22.9 min (major), 73% ee.

(*S*)-3-(Ferrocenyl)-1-phenylpentan-1-one **13m**:¹⁷ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 14.5 min (major), 26.6 min, 71% ee.

(*R*)-1-(4-Chlorophenyl)-3-phenylpentan-1-one **13n**:¹⁶ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 13.1 min, 16.6 min (major), 82% ee.

(*R*)-1-(4-Methoxyphenyl)-3-phenylpentan-1-one **13o**:¹⁶ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 22.6 min, 37.3 min (major), 65% ee.

(*R*)-1-(Ferrocenyl)-3-phenylpentan-1-one **13p**: The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 16.7 min, 21.4 min (major), 53% ee.

(*R*)-1,3-Bis(4-chlorophenyl)pentan-1-one **13q**:¹² The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 14.0 min, 21.4 min (major), 91% ee.

(*R*)-1-(4-Chlorophenyl)-3-(4-methoxyphenyl)pentan-1-one **13r**: The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 17.5 min, 28.2 min (major), 71% ee.

(*S*)-1-(4-Chlorophenyl)-3-ferrocenylpentan-1-one **13s**: The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 14.7 min (major), 38.0 min, 87% ee.

(*R*)-3-(4-Chlorophenyl)-1-(4-methoxyphenyl)pentan-1-one **13t**: The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 11.7 min, 25.0 min (major), 84% ee.

(*R*)-1,3-Bis(4-methoxyphenyl)pentan-1-one **13u**:²⁰ The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and detected at a UV wave length of 254 nm. Retention times: 16.1 min, 36.1 min (major), 39% ee.

(*S*)-1-(4-Methoxyphenyl)-3-ferrocenylpentan-1-one **13v**: The enantiomeric excess was determined on a Daicel Chiralpak AD-H column with hexane/2-propanol = 95/5, flow = 0.5 mL/min, and

detected at a UV wave length of 254 nm. Retention times: 12.9 min (major), 49.6 min, 39% ee.

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Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.tetasy.2016.01.016>.

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